Interpretable Gene Programs or Foundation Models?

Comparing Sparse Dictionary Learning and scGPT
for Single-Cell Classification

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Abstract

We compare processed gene expression, sparse dictionary codes, and frozen scGPT embeddings for human pancreatic cell-type classification using the same regularized multinomial logistic model family. On a fixed stratified split of 8,506 Baron cells, test Macro-F1 is 0.9939 for expression, 0.9810 for a 64-atom dictionary, and 0.9624 for a small custom scGPT checkpoint. A separate retrospective nested donor-held-out analysis matches gene availability, standardizes classifier inputs, and varies dictionary initialization. Its respective donor-mean Macro-F1 scores are 0.9246, 0.9215, and 0.8823. Dictionary predictions vary little across the tested initializations, whereas matched atoms are only moderately reproducible. We relate these observations to sparse-coding optimality, factor nonidentifiability, and the effect of scaling on regularization. The results support sparse compression and the importance of a strong expression baseline within this study; they do not establish biological pathway recovery or general superiority over pretrained single-cell models.

1 Introduction and related work

Single-cell RNA sequencing represents each cell through thousands of gene measurements. A representation can be useful because it improves prediction, compresses the data, or exposes patterns that can be inspected. These are separate goals: an accurate classifier need not reveal stable biological programs, and low reconstruction error need not imply useful cell-type information. We ask whether a frozen pretrained representation provides more linearly accessible cell-type information than a sparse representation learned from the target training cells, and whether any compression is accompanied by reproducible gene-level structure.

We compare three feature sets under the same logistic model family: processed expression, signed sparse dictionary codes, and frozen scGPT embeddings. The expression baseline tests whether representation learning is necessary for this task. Our contributions are a provenance-checked comparison with frozen test evaluation; a retrospective donor, scaling, and initialization analysis; and an explicit connection between the optimization objectives, numerical certificates, and the limits of interpretation. The mathematics is established optimization theory applied to this experiment, not a claim of new theorems.

Sparse coding learns reusable components with sparse activities [2]; fixing the dictionary yields an ℓ_1-penalized regression problem [1]. Online dictionary learning provides an established approach to the joint factorization [3], while proximal methods explain the thresholding behavior of its convex coding subproblem [4]. For single-cell data, consensus nonnegative matrix factorization has been used to study identity and activity programs [5]. Our signed atoms differ from nonnegative programs and require sign/permutation alignment before stability comparisons.

scGPT motivates pretrained representations for single-cell tasks [6]; work on pretraining size and diversity motivates examining which benefits transfer to downstream evaluation [7]. Our comparison uses a small custom checkpoint trained by the project team, not the original published large scGPT weights. Conclusions therefore concern this checkpoint, this cohort, and regularized linear probing.

2 Data and methods

2.1 Data and shared cohort

We use the human subset of the Baron pancreas dataset (GSE84133) [8], obtained through Bioconductor's `scRNAseq::BaronPancreasData` [9]. The prepared raw-count matrix has 8,569 cells, 20,125 genes, four donor IDs, and 14 supplied cell-type annotations. Conversion validates finite nonnegative integer counts and preserves unique cell IDs, donor IDs, and labels in AnnData. Labels are taken as given rather than independently reannotated.

The shared policy removes classes with fewer than 30 cells: T cells (7), Schwann (13), epsilon (18), and mast (25), leaving 8,506 cells in ten classes. No additional cells fail the matched scGPT-panel check. This cutoff permits representation of every retained class in each partition, but narrows the task and does not solve within-donor class imbalance. A fixed seed-42, cell-type-stratified 70/15/15 split yields 5,954 training, 1,277 validation, and 1,275 test cells after integer rounding. Donors overlap across these partitions. Appendix C gives donor/class counts.

2.2 Expression and dictionary features

Classical preprocessing divides each cell by its total count across all original genes, multiplies by 10,000, and applies $\log(1+x)$. In the original experiment, the 2,000 genes with greatest training log-expression variance are retained; this is variance ranking, not dispersion-adjusted highly variable gene selection. Gene-wise training means and population standard deviations define z-scores, with no clipping. Held-out cells reuse these parameters and gene ordering. The baseline uses these features directly.

For $X \in \mathbb{R}^{n\times p}$, we learn signed codes $Z \in \mathbb{R}^{n\times K}$ and signed atoms $D \in \mathbb{R}^{K\times p}$:

$$\min_{Z,D} \ \frac{1}{n}\left(\frac{1}{2}\|X - ZD\|_F^2 + \alpha\|Z\|_1\right), \qquad \|d_k\|_2 \le 1. \tag{1}$$

The norm constraint prevents $(Z/c, cD)$ from preserving reconstruction while arbitrarily reducing the penalty. Fixed-D coding is convex; jointly learning both factors is not. We use scikit-learn mini-batch dictionary learning with coordinate-descent fitting and penalized Lasso encoding [10]. Both use the same α. Fitting uses float64, batch size 128, one numerical thread, a ten-epoch ceiling, tolerance 0.001, and a ten-batch no-improvement stopping heuristic. Encoding allows 3,000 iterations. Early stopping is not an optimality certificate. Held-out encoding never updates the dictionary.

2.3 Frozen scGPT features

The fixed checkpoint, `sctab_seed0_n1600000_scgpt`, has 1,166,466 parameters and was pretrained on 1.6 million sampled cells for ten epochs. The recorded choice uses the largest completed cell-number condition, sampling seed 0, and its final epoch; Baron classification scores were not used to select it. Pretraining overlap with Baron has not been audited.

The mapped raw input matches 1,879 of 2,000 fixed panel genes. The remaining 121 columns are zero-filled; extraction uses nonzero panel counts as gene tokens, prepends CLS, bins expression into 51 bins, pads/truncates to 1,200 tokens, and performs no masked prediction or random token sampling. The frozen encoder returns L2-normalized 128-dimensional CLS vectors. CPU float32 extraction used batch size 8 and preserved all 8,569 IDs; downstream joins select the shared cohort by ID. This processing differs from classical log normalization and z-scoring.

2.4 Common classifier and regularization geometry

Each representation receives a separate multinomial logistic classifier with unpenalized intercept, L2-penalized weights, and no class weighting. For input u_i, weights W, and intercept b, the mean-loss convention is

$$\frac{1}{n}\sum_i \left[-(u_i W + b)_{y_i} + \log \sum_c e^{(u_i W + b)_c}\right] + \frac{\lambda}{2}\|W\|_F^2. \tag{2}$$

The implementation uses $C = 1/(n\lambda)$ [10]. In the original experiment, features receive no additional classifier-side scaling. In the donor study, $u = (h - m)S^{-1}$ uses training-fitted coordinate means and scales for all methods. In original coordinates $B = S^{-1}W$, the penalty becomes $\lambda\|SB\|_F^2/2$: a single tuned scalar C cannot undo arbitrary coordinate-wise scaling. Scaling is therefore part of the statistical model, not merely an optimization convenience. Appendix A supplies derivations and qualifications.

3 Experimental design

3.1 Original stratified-cell comparison

A starter dictionary grid used $K \in \{16, 32\}$ and $\alpha \in \{1, 5\}$. One validation-driven refinement used $K \in \{32, 64\}$, $\alpha \in \{0.5, 1, 2\}$, and seed 42. This was an adaptive, modest search, not a fully prespecified search over all dictionary choices. Each of the six refined dictionaries, expression, and scGPT received the same classifier search: coarse $C \in \{0.01, 0.1, 1, 10, 100, 1000\}$, then 21 logarithmically spaced values within a factor of ten of the coarse winner, excluding values already tested. All 192 classifier fits in the refinement converged.

Validation Macro-F1 selects C, with ties preferring smaller C. Dictionary ties prefer fewer atoms, then fewer active validation coefficients, then candidate ID. The selected dictionary has $K = 64, \alpha = 2$. Selected C values are 0.316228 (expression), 0.063096 (dictionary), and 316.227766 (scGPT). The LBFGS classifier uses at most 5,000 iterations, tolerance 10^{-4}, and one numerical thread. Coefficients and selection records are frozen before test evaluation, with no training-plus-validation refit. Test predictions are generated once from these saved models; later robustness analysis leaves that experiment unchanged.

3.2 Retrospective nested donor evaluation

Each of four outer folds holds out one donor. Among the other three donors, each serves once as inner validation while the other two provide training. All classical transforms, dictionaries, and classifier scalers are refitted inside each fold. Classical methods use the 1,879 measured fixed-panel genes rather than training-selected 2,000 genes, while library totals still use all raw genes. This controls measured gene availability, not the distinct upstream processing or pretraining of scGPT.

The primary protocol standardizes every classifier input. An unstandardized ablation is evaluated separately with its own inner tuning, not selected by outer performance. Inner tuning uses

$\lambda \in \{10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$ in Equation (2), with ties preferring stronger regularization. Candidates must converge in every inner fit. Dictionary scores average seeds 0, 1, and 2 as well as inner donors. Initial atoms are normalized training-cell vectors; only their selection varies, with mini-batch random state fixed at 42. The classifier tolerance is 10^{-5}.

The cohort, $K = 64$, and $\alpha = 2$ are inherited from earlier work, not nested-selected anew. All donors contributed to previously inspected results; this analysis is retrospective and cannot provide an untouched confirmatory test. An earlier sensitivity run varied the estimator random state, mixing initialization and batch randomness. It is retained as exploratory provenance; the controlled analysis separates these effects and averages all seeds.

3.3 Metrics, numerical accuracy, and aggregation

Macro-F1 is the unweighted mean of class F1 scores; balanced accuracy is mean class recall. We also report accuracy and mean multiclass negative log probability. Class order, labels, and cell IDs are aligned across methods. For donor results, dictionary seeds are averaged within donor before taking equal-donor means and descriptive sample SDs. These are not confidence intervals: only four donors are available, outer training folds overlap, and random seeds are not biological replicates.

For fixed-D coding, $g = (zD - x)D^\top$ must satisfy $g_k = -\alpha \operatorname{sign}(z_k)$ when $z_k \neq 0$ and $|g_k| \leq \alpha$ otherwise. These KKT conditions certify the convex coding subproblem, not the learned dictionary. Positive $\lambda_{\min}(DD^\top)$ establishes conditional coding uniqueness. We also measure the projected dictionary-gradient residual for the mean reconstruction loss. Atom stability uses maximum-total absolute-cosine assignment between same-fold dictionaries, sign-aligned training-code correlations, and top-20 absolute-weight gene Jaccard overlap. This handles sign and permutation but not program splitting or merging.

4 Results, discussion, limitations, and conclusion

4.1 Classification and sparse compression

Expression has the strongest observed original test performance on all four metrics (Table 1). The dictionary compresses 2,000 input genes to 64 coordinates, a 31.25-fold dimensional reduction, while retaining high accuracy. Its small validation Macro-F1 advantage (0.984580 versus 0.984365) does not persist on test. This is not evidence of superiority or equivalence; we do not retune using test outcomes.

Table 1: Original frozen evaluation on 1,275 test cells from the shared donor pool. Higher F1/accuracy is better; lower log loss is better.

Representation	Macro-F1	Accuracy	Balanced acc.	Log loss
Expression	0.993913	0.994510	0.993687	0.019454
Dictionary	0.980962	0.989020	0.977933	0.036989
scGPT	0.962358	0.968627	0.955637	0.096535

The selected dictionary has no unused atoms in training or validation. With activity defined as $|z_k| > 10^{-6}$, validation codes are 17.02% active (10.89 atoms per cell); training codes are 15.30% active. Validation reconstruction MSE is 0.7826, or 0.7924 of the zero-reconstruction MSE. This is a reconstruction diagnostic, not a classification result. Lower penalties improve reconstruction

in this grid without monotonically improving classification (Appendix C). The observed baseline strength shows why comparison against expression itself is necessary.

4.2 Donor separation and classifier scaling

Table 2 summarizes the retrospective matched-panel study. Expression and dictionary donor-mean Macro-F1 are close, while scGPT is lower. Four donors and overlapping training folds do not support a broad significance or equivalence claim. These values cannot be subtracted from the original test scores to estimate a pure donor effect: grouping, gene policy, scaling, regularization selection, and dictionary initialization also changed. The scale ablation changes scGPT Macro-F1 substantially,

Table 2: Retrospective donor Macro-F1. Standardization is the primary protocol; SD describes four donor scores after averaging dictionary seeds within donor.

Representation	Standardized mean ± SD	Unscaled mean
Expression	0.9246 ± 0.0549	0.9246
Dictionary	0.9215 ± 0.0488	0.9220
scGPT	0.8823 ± 0.0473	0.8440

despite using the same classifier family and separately tuning regularization. Dictionary Macro-F1 changes little here; expression is already standardized upstream. This supports treating feature scale as part of a fair linear-probing protocol, without claiming that matched genes and scaling eliminate all representation differences.

4.3 Numerical accuracy and interpretation

Controlled dictionary initialization yields within-donor Macro-F1 SDs of 0.0008–0.0030. Atom reproducibility is weaker: mean matched absolute cosine is 0.606, top-20-gene Jaccard overlap 0.344, and sign-aligned training-code correlation 0.572. Stable prediction therefore does not imply uniquely recovered gene programs.

Across 48 donor-study dictionary fits, coding KKT residuals are at most 1.1×10^{-6} and the smallest observed Gram eigenvalue is about 0.00488. These support accurate, conditionally unique encodings. Projected dictionary-gradient norms remain 1.90–4.06 and early stopping occurs after 2–7 epochs: absence of warnings does not establish joint stationarity or global optimality. All 640 donor-study classifier fits completed without convergence warnings, and the 40 saved outer prediction sets were checked against IDs, labels, and recomputed metrics.

The original fitted dictionary is directly inspectable. For example, atom 2 has large positive weights on CUZD1, CTRC, and PNLIPRP1 and elevated mean activity in acinar cells; atom 5 emphasizes COL1A2, COL3A1, and COL1A1 and is active in annotated activated stellate cells. These are observed weight/activity associations (Appendix D), not formal enrichment or biological validation. Signs and indices are run-specific. No causal pathway claim follows, particularly given moderate atom stability.

4.4 Limitations and conclusion

The original split shares donors, and the secondary study is retrospective, uses only four donors from one study, and inherits earlier choices. Some held-out donors contain only two or three acinar cells; rare-class exclusion further restricts the estimand. Annotation quality is assumed.

The small custom scGPT model, missing panel genes, distinct input transforms, and unaudited pretraining overlap limit attribution to architecture or pretraining. The adaptive validation search and three initialization seeds provide limited coverage. Formal pathway enrichment, external-dataset confirmation, and comprehensive hyperparameter or seed sweeps were not done.

For this cohort and linear-probing protocol, the supplied scGPT representation does not improve over processed expression or sparse dictionary codes. Dictionary learning offers sparse compression and inspectable associations, but not established biological identifiability. The strongest methodological lesson is to distinguish coding accuracy, predictive stability, and biological reproducibility, and to control both donor separation and regularization geometry. A future confirmatory study requires an untouched cohort, a protocol fixed in advance, and a pretraining-overlap audit.

A Theory and optimization details

These derivations consolidate the design note and mathematical addendum. They apply established convex and constrained optimization facts to the implemented estimator; they do not assert a new recovery theorem.

A.1 Sparse coding: optimality, uniqueness, and diagnostics

For a row $x \in \mathbb{R}^p$ and a fixed $D \in \mathbb{R}^{K\times p}$, define

$$f_x(z) = \tfrac{1}{2}\|x - zD\|_2^2 + \alpha\|z\|_1, \qquad \alpha > 0.$$

Writing $g = (zD-x)D^\top$, convex subgradient optimality gives the necessary and sufficient conditions

$$g_k = -\alpha \operatorname{sign}(z_k) \quad (z_k \neq 0), \qquad |g_k| \leq \alpha \quad (z_k = 0).$$

Consequently $z = 0$ is optimal exactly when $\|xD^\top\|_\infty \leq \alpha$. The executable diagnostics use the coordinate residual

$$r_k = \begin{cases} |g_k + \alpha \operatorname{sign}(z_k)|, & z_k \neq 0, \\ \max(|g_k| - \alpha, 0), & z_k = 0. \end{cases}$$

Exact nonzero coefficients are used for this optimality diagnostic, distinct from the 10^{-6} activity threshold used for sparsity summaries. Small residuals certify approximate coding optimality numerically, not statistical recovery of a true gene program.

Proposition 1 (Conditional uniqueness and error certificate)**.** *If $\mu = \lambda_{\min}(DD^\top) > 0$, f_x is μ-strongly convex and has a unique minimizer z^*. For any subgradient $v \in \partial f_x(z)$,*

$$\|z - z^*\|_2 \leq \frac{\|v\|_2}{\mu}, \qquad f_x(z) - f_x(z^*) \leq \frac{\|v\|_2^2}{2\mu}.$$

Proof. The quadratic Hessian is $DD^\top \succeq \mu I$. Adding a convex L1 term preserves strong convexity. Strong monotonicity with $0 \in \partial f_x(z^*)$ gives $\mu\|z - z^*\|^2 \leq \langle v, z - z^*\rangle$; apply Cauchy–Schwarz. The strong subgradient inequality at z gives $f_x(z) - f_x(z^*) \leq \langle v, z - z^*\rangle - \mu\|z - z^*\|^2/2$, whose maximum over the displacement is $\|v\|^2/(2\mu)$. □

The coordinatewise least-norm subgradient has absolute coordinates r_k. A reported infinity residual yields $\|v\|_2 \leq \sqrt{K}\|r\|_\infty$. The bounds require positive μ and can be uninformative for an ill-conditioned dictionary. They do not imply uniqueness of the jointly learned factors.

Proposition 2 (What increasing the penalty actually guarantees)**.** *Fix D, x and let z_j minimize $E(z) + \alpha_j L(z)$, where $E = \|x - zD\|^2/2$, $L = \|z\|_1$, and $0 < \alpha_1 < \alpha_2$. Then $L(z_2) \leq L(z_1)$ and $E(z_2) \geq E(z_1)$.*

Proof. The two optimality inequalities are $E_1 + \alpha_1 L_1 \leq E_2 + \alpha_1 L_2$ and $E_2 + \alpha_2 L_2 \leq E_1 + \alpha_2 L_1$. Adding gives $(\alpha_2 - \alpha_1)(L_2 - L_1) \leq 0$. Substitute into the first inequality to obtain the reconstruction statement. □

This does not establish monotonic support cardinality for correlated atoms, nor a monotonic relation across independently learned dictionaries. Predictive accuracy need not be monotone in either quantity.

A.2 Joint optimization and identifiability

For $X \in \mathbb{R}^{n\times p}$, use

$$\min_{Z,D} \ \frac{1}{n}\left(\frac{1}{2}\|X - ZD\|_F^2 + \alpha\|Z\|_1\right), \qquad \|d_k\|_2 \le 1.$$

Dividing the entire objective by n leaves minimizers unchanged. Without the norm constraint $(Z/c, cD)$ preserves reconstruction while reducing the penalty. Conditional subproblems are convex, but joint optimization is nonconvex. Mini-batch early stopping does not prove stationarity or global optimality. In addition to coding KKT residuals, record

$$\|D - \Pi_{\mathcal{D}}(D - G)\|_F, \qquad G = Z^\top(ZD - X)/n,$$

where projection normalizes only rows exceeding unit norm. A zero projected-gradient residual is first-order optimality for the fixed-Z constrained problem; a nonzero value measures a remaining dictionary-update residual, not a bound on global suboptimality. The implementation is based on established online dictionary methods, but we do not transfer their asymptotic theorems to our finite solver run without verifying assumptions [3].

Permutation and simultaneous sign changes preserve the objective. For two dictionaries in the *same fold and gene coordinates*, compare normalized atom similarities $S_{ij} = |\langle d_i, d'_j\rangle|/(\|d_i\|\|d'_j\|)$ by solving

$$\max_{\pi\in\mathfrak{S}_K} \sum_{i=1}^{K} S_{i,\pi(i)}.$$

We use linear assignment, align signs, and report matched cosines, top-20 absolute-weight gene Jaccard overlaps, and training-code correlations. This one-to-one metric can penalize program splitting/merging; it is not a complete equivalence test. Seeds are repeated computations, not independent biological samples. Controlled runs vary normalized training-cell initial atoms while fixing mini-batch random state 42; no best seed is selected.

A.3 Scaling changes regularization geometry

For row features h and training-fitted center m and diagonal scale S, use $u = (h - m)S^{-1}$ for *every* representation. With class weights W and an unpenalized intercept b, minimize

$$\frac{1}{n}\sum_i \left[-(u_iW + b)_{y_i} + \log\sum_c e^{(u_iW+b)_c}\right] + \frac{\lambda}{2}\|W\|_F^2.$$

Our installed sklearn implementation uses $C = 1/(n\lambda)$ for this mean-loss convention [10]. This maintains the same regularization definition when inner and outer training sample sizes differ. Using a fixed C instead corresponds to a different λ as n changes.

If original-coordinate weights are $B = S^{-1}W$, the penalty is $\lambda\|SB\|_F^2/2$. Thus normalization is part of the statistical model: tuning a single scalar C cannot undo arbitrary coordinate-wise rescaling. With positive scales, standardization removes arbitrary affine changes in feature units (up to intercept adjustment). It does not make representation dimensions, information content or intrinsic geometry identical. Strong convexity in penalized weight directions does not remove the softmax common-intercept gauge; do not claim unrestricted uniqueness of (W, b).

For the matched-gene secondary experiment, expression and dictionary use the 1,879 measured genes from the fixed 2,000-gene scGPT panel. Classical per-cell library normalization still uses all raw genes, followed by log1p and training centering/scaling. scGPT retains its required panel processing and pretrained nonlinear transform; identical biological features do not imply identical

preprocessing. This is a matched-gene control, not proof that all confounding disappears. Standardization is primary; an unstandardized ablation is reported without selecting between them using outer scores.

A.4 Nested grouped evaluation and aggregation

For each outer donor d, restrict all fitting and tuning to the other three donors. Each of those three serves once as inner validation while the other two supply training. Refit classical preprocessing, dictionary and classifier scaler inside every inner fold. Choose λ by mean inner-donor Macro-F1; dictionary scores also average the three predeclared initialization seeds. Ties prefer stronger regularization, and a candidate must converge in every inner fit. Then refit on the three outer training donors and evaluate only donor d.

Report all donor scores. Average dictionary seeds *within donor* before computing the equal-donor mean and donor standard deviation. Donors have unequal cell counts; pooling cells targets a different weighting. Do not treat three seeds times four donors as twelve independent biological replicates or use a cell-level bootstrap to claim donor-level uncertainty. Outer training sets overlap; four donors provide limited evidence even with group separation. Learned hyperparameters K, α and the cohort remain inherited exploratory choices, not nested-selected choices in this secondary experiment.

A.5 Illustrative updates and implementation distinction

For illustration, a proximal-gradient update is

$$z^{(t+1)} = S_{\eta\alpha}\Big(z^{(t)} - \eta(z^{(t)}D - x_i)D^\top\Big),$$

where $S_a(u) = \operatorname{sign}(u)\max(|u|-a, 0)$ acts entrywise. For nonzero D, a step size $0 < \eta \le 1/\|DD^\top\|_2$ is a standard conservative choice. Proximal methods and their accelerated variants are developed by Beck and Teboulle [4]. The update above is the unaccelerated step, not the full FISTA algorithm. This explains the role of soft thresholding; the reported estimator uses coordinate descent for this subproblem.

Dictionary updates. With Z fixed, updating D is a convex constrained least-squares problem. A projected-gradient step takes

$$\widetilde{D} = D - \eta_D Z^\top(ZD - X), \qquad d_k^{\text{new}} = \frac{\widetilde{d}_k}{\max(1, \|\widetilde{d}_k\|_2)}.$$

An appropriate step size is needed; arbitrary gradient steps followed by projection do not guarantee improvement. Alternating sparse-code and dictionary updates exploits the blockwise convex structure, but the joint problem is nonconvex and initialization matters. Exact block minimization cannot increase the objective, yet this does not establish convergence to a global optimum. Mini-batch or approximate updates require their own convergence diagnostics.

The proximal and projected steps explain the objectives; the reported runs use the library mini-batch/coordinate-descent solver, not a custom FISTA implementation. The projected step above uses the summed loss; dividing its gradient by n gives the mean-loss convention used by our diagnostic.

B Experimental workflow and reproducibility

B.1 Resources and artifact contract

The root README documents installation, resource destinations, and all CLI parameters. Exact offline reproduction requires the raw `data/processed/baron_raw.h5ad`, the final split directory's `baron_split_manifest.csv`, `split_settings.json`, and `baron_exclusions.csv`, and four files under `code/scgpt/pretraining_handoff/outputs/baron/`: `baron_scgpt_full.npz`, `baron_scgpt_full.metadata.json`, `checkpoint_selection.json`, and `preparation_report.json`. These large/local artifacts are not in Git; obtain the compatible bundle from the authors. A public complete-bundle URL is not yet documented. The optional weight/sampling download and extraction environment are specified in the scGPT handoff README. Classification from full embeddings does not require weights or a GPU.

Raw counts are preserved in AnnData together with cell types, donors, and gene symbols. Final manifests contain `cell_id, cell_type, donor, split`. Classical NPZ files contain features, cell IDs, labels, donors, and ordered gene IDs; fitted preprocessing stores selected columns and training statistics. Dictionary outputs store atoms, ordered genes, training/validation codes, parameters, and diagnostics. Classifier NPZ files contain coefficients, intercepts, and class order; JSON records bind source and output hashes. Numeric/string arrays are loaded without object pickling. Joins reject duplicate or missing IDs, nonfinite features, mismatched labels, and gene-order changes. Different feature dimensions are intentional; representations are not concatenated.

B.2 Execution order

Run the following from the repository root in the project environment. Reusing the exact supplied raw/split bundle is preferable for matching the recorded extraction hashes. A fresh public-data build may serialize differently; inspect provenance differences rather than replacing hashes to bypass checks.

```
python3 -m venv .venv
source .venv/bin/activate
python -m pip install -r requirements.txt
# Public-data route: Rscript and a writable R library are required.
python scripts/download_data.py
python scripts/preprocess_data.py --scgpt-confirmed
# Supply the four compatible embedding files before this check.
python scripts/load_scgpt.py
# Optional execution check; no model selection or test scoring:
python scripts/run_end_to_end.py --check-only
python scripts/run_end_to_end.py --plots
# Full validation selection in a fresh output directory:
python scripts/tune_models.py --output results/workflow/reproduction
```

If classical features already exist, skip their creation. Downloading and preprocessing are unnecessary for the smoke route when the eight exact inputs are supplied. For full selection, review `recommended_models.csv` and use its candidate/C choices with the generated `validation_config.json`. For each method, train with `scripts/train_logistic.py`, evaluate with `scripts/evaluate_validation.py--freeze`, and then use `scripts/run_test.py` with that matching selection file. The root README provides complete arguments. No preprocessing or classifier refit on the training-plus-validation union is used in the reported experiment.

```
python scripts/run_donor_study.py \
  --output results/workflow/donor_reproduction
python scripts/summarize_donor_study.py \
  --study results/workflow/donor_reproduction --plots
```

The donor study refits folds from raw data rather than reusing the original training-feature bundle. Output paths must be fresh. Preserve frozen models and their referenced inputs; ignored local artifacts are not transferred by a Git clone. Test scores must not drive further tuning.

B.3 Versions, provenance, and report construction

The donor protocol records Python 3.14.6, NumPy 2.5.3, SciPy 1.18.1, scikit-learn 1.9.1, and pandas 3.0.6. The separate CPU embedding extraction records PyTorch 2.11.0+cu128 and NumPy 2.2.6, float32, four threads, batch size 8, and seed 2026092500. These are recorded environments, not a claim that an unpinned dependency installation gives identical binaries or outputs. The extraction metadata does not identify a CPU model; no hardware-normalized runtime comparison is made.

The raw AnnData SHA-256 is `452177f01994c2f81ad79cb31b1aaf562b12ff4d56d52299dce1f5ed18c08176`; the shared manifest hash is `bb1e6f0810e20d02400892743c49077323d30bb67e57e043387da84115affca5`; the embedding archive hash is `08ee0912bc6f43f3ecc79aed1f4bdc4683587f47e3e47547be282aa5927f47a6`. Full source, input, and output hashes are in the run reports. The preserved pre-test freeze manifest describes its original pre-test state; the later test evaluation report separately records completion. Model files were not changed by evaluation. No single repository commit is asserted to describe all runs; recorded script hashes identify the executed sources.

The table/figure snapshots included here are generated from saved artifacts by `scripts/build_report_assets.py`. Its manifest records source and snapshot hashes. Rebuilding assets requires the local experiment directories; compiling this report requires only the tracked snapshots. From `docs/notes`, compile `main.tex` twice with pdfLaTeX, or upload that directory to Overleaf and choose `main.tex` as the main document. The root assembles six section files; the eight-page main-report budget excludes these appendices and references. The pre-consolidation notes are preserved locally in `backups/notes_before_consolidation_*.zip`.

C Supplementary tables

Table 3: Retained cell-type counts by donor. Headers abbreviate GSM2230757–GSM2230760.

Cell type	0757	0758	0759	0760
acinar	110	3	843	2
activated_stellate	51	81	100	52
alpha	236	676	1130	284
beta	872	371	787	495
delta	214	125	161	101
ductal	120	301	376	280
endothelial	130	23	92	7
gamma	70	86	36	63
macrophage	14	17	14	10
quiescent_stellate	92	22	54	5

Table 4: Refined dictionary grid: validation-selected classifier C, validation Macro-F1, active coefficient percentage, and MSE relative to zero reconstruction. These selection scores are not independent test estimates.

K	α	C	Macro-F1	Active (%)	Relative MSE
64	2.0	0.0631	0.984580	17.02	0.7924
32	1.0	0.0631	0.982813	56.46	0.7870
64	0.5	0.05012	0.981766	71.51	0.7622
32	0.5	0.0631	0.981761	79.73	0.7779
64	1.0	0.631	0.979681	42.63	0.7755
32	2.0	1.995	0.972629	26.23	0.8033

Table 5: Retrospective standardized outer-donor Macro-F1. Dictionary mean and sample SD are over its three initializations within that donor.

Donor	Expression	Dictionary	Seed SD	scGPT
GSM2230757	0.980395	0.974973	0.003016	0.949713
GSM2230758	0.889170	0.886383	0.000831	0.877546
GSM2230759	0.961553	0.950132	0.002520	0.860728
GSM2230760	0.867143	0.874391	0.002737	0.841405

Table 6: Original frozen test per-class F1 and cell counts. Small class support limits interpretation of perfect scores.

Cell type	Cells	Expression	Dictionary	scGPT
acinar	144	0.9930	0.9931	0.9724
activated_stellate	42	1.0000	0.9655	0.9762
alpha	349	0.9957	0.9957	0.9730
beta	379	0.9947	0.9921	0.9725
delta	90	0.9945	0.9721	0.9432
ductal	161	0.9938	0.9875	0.9781
endothelial	38	1.0000	1.0000	0.9610
gamma	38	0.9870	0.9870	0.8857
macrophage	8	1.0000	1.0000	1.0000
quiescent_stellate	26	0.9804	0.9167	0.9615

Table 7: Illustrative original-dictionary associations: leading positive genes and cell-type mean codes standardized using training statistics. Atom numbers are one-based and run-specific; no enrichment test was performed.

Atom	Positive genes	Cell type	Train	Validation
2	CUZD1, CTRC, PNLIPRP1	acinar	2.442	2.265
5	COL1A2, COL3A1, COL1A1	activated_stellate	4.524	4.480
11	PLVAP, STC1, ENG	endothelial	5.522	5.425

D Supplementary figures

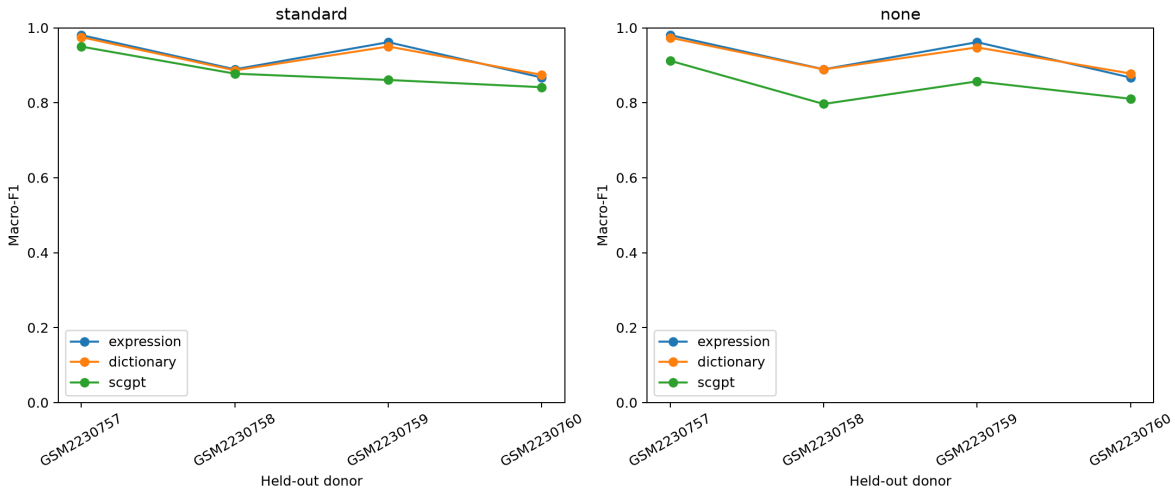


Figure 1: Retrospective donor Macro-F1 by representation and classifier scaling. Dictionary scores average initialization seeds within donor; connecting donors does not imply independent repeated experiments or a temporal ordering.

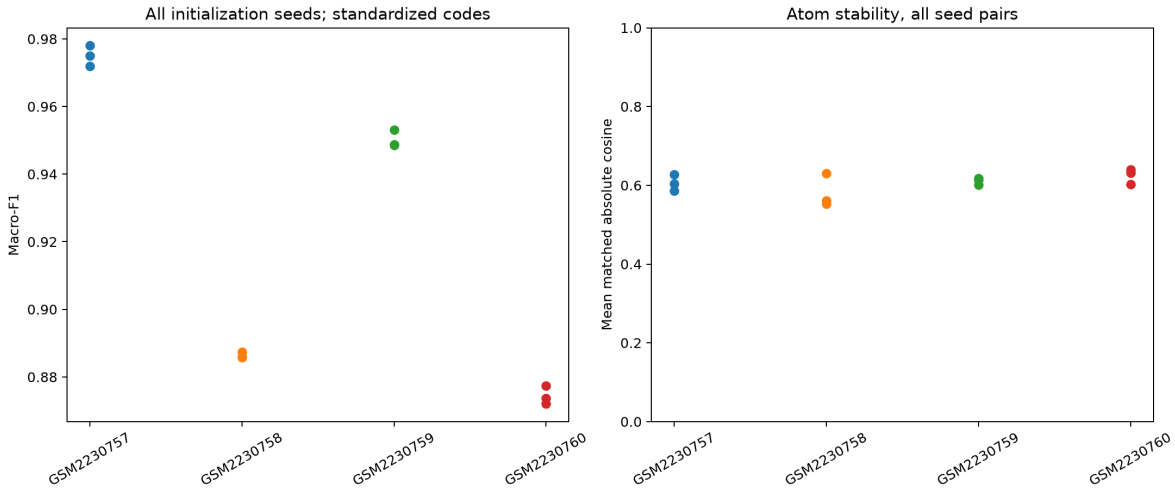


Figure 2: Controlled initialization diagnostics. Prediction stability and matched-atom similarity measure distinct properties. Seed pairs are repeated computations on the same training folds, not biological replicates.

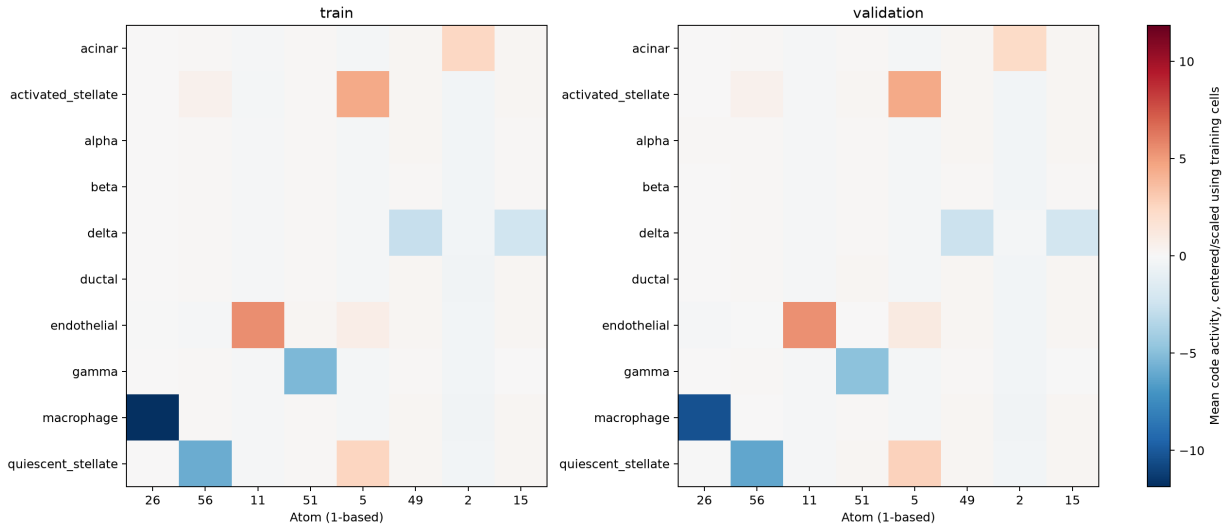


Figure 3: Original selected dictionary: signed cell-type mean activities, standardized using training-code statistics. Eight displayed atoms are selected by training-only cell-type contrast; validation uses the same atoms and scale. This descriptive visualization is not a pathway-enrichment test.

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